Simple linear regression

- The sample: $\{(x_i, y_i)\}_{i=1}^{n}$

independent variable

dependent variable

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

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Multiple linear regression with k independent variables. k>1

- The sample: $\{(x_{1i}, x_{2i}, \ldots, x_{ki}, y_i)\}_{i=1}^{n}$

independent variables

dependent variable

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k}_{\text{True regression line}} + \varepsilon$$

$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

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Various Multiple Linear Regression Models

- **Polynomial regression model** (the independent variables are all powers of a single variable).

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + \varepsilon_i$$

- **Quadratic model** (polynomial regression of model of degree 2, and powers of several variables).

$$y_i = \beta_0 + \beta_1 x_{1_i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \beta_3 x_{1i}^2 + \beta_3 x_{2i}^2 + \varepsilon_i$$

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Various Multiple Linear Regression Models

- A variable that is the product of two other variables is called an **interaction**.
- These models are considered linear models, even though they contain nonlinear terms in the independent variables. The reason is that *they are linear in the coefficients,* β_i.

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Estimating the Coefficients

- In any multiple regression model, the estimates

$$\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$$

are computed by least-squares, just as in simple linear regression. The equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$$

is called the **least-squares equation** or **fitted regression equation**.

- Now define $\hat{y}_i$ to be the *y* coordinate of the least-squares equation corresponding to the *x* values $(x_{1i}, \ldots, x_{pi})$.

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Estimating the Coefficients

- The residuals are the quantities.

$$e_i = y_i - \hat{y}_i$$

which are the differences between the observed *y* values and the *y* values given by the equation.

- We want to compute $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ so as to minimize the sum of the squared residuals. The formulas are complicated, so we rely on computers to calculate them.

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Multiple regression with k independent variables. k>1:

Source	df	Sum Sq.	Mean Sum Sq.	F
Regression	k	SSR	$MSR = \frac{SSR}{k}$	$\frac{MSR}{MSE}$
Error	n-(k+1)	SSE	$MSE = S^2 = \frac{SSE}{n-(k+1)}$	
Total	n-1	SST		

When all β_i=0:

$$MSR = \frac{SSR}{k} \sim \chi^2_k$$

$$MSE = S^2 = \frac{SSE}{n-(k+1)} \sim \chi^2_{n-(k+1)}$$

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ANalysis Of VAriance identity

Let all β_i=0.

k: number of independent variables

$$MSE = S^2 = \frac{SSE}{n-(k+1)} \sim \chi^2_{n-(k+1)}$$

$$MSR = \frac{SSR}{k} \sim \chi^2_k$$

What is the distribution of $\frac{SSR/k}{SSE/n-(k+1)}$?

$$\frac{W/u}{Y/v} \sim F_{u,v} \qquad W \sim \chi^2_u, \; Y \sim \chi^2_v$$

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Multiple regression with k independent variables. k>1:

Source	*df*	*Sum Sq.*	*Mean Sum Sq.*	*F*
Regression	*k*	*SSR*	$MSR = \frac{SSR}{k}$	$\frac{MSR}{MSE}$
Error	*n-(k+1)*	*SSE*	$MSE = S^2 = \frac{SSE}{n-(k+1)}$	
Total	*n-1*	*SST*		

$H_0: \beta_i = 0$ for all i
$H_a: \beta_i \neq 0$ for at least one i

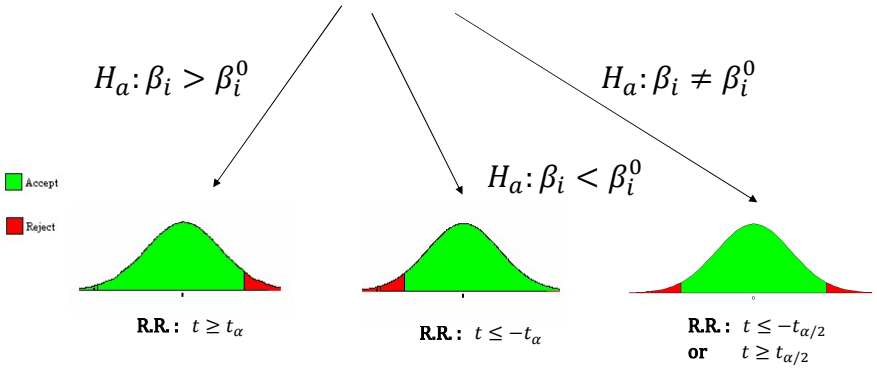
can be tested via: → reject H_0 if $F > F_{\alpha,k,n-(k+1)}$.

Overall significance test

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Individual Significance Test
Hypothesis testing on β_i

$H_0: \beta_i = \beta_i^0$ **Test statistics:** $T = \frac{b_i - \beta_i^0}{s_{b_i}} \sim t_{n-(k+1)}$



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Example: Water Consumption

A production plant cost-control engineer is responsible for cost reduction. One of the costly items in his plant is the amount of water used by the production facilities each month. He decided to investigate water usage by collecting 17 observations on his plant's water usage and other variables. All his data as well as MINITAB outputs corresponding to several regression models are given below.

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Definition of variables

X_1 = daily amount of production (in pounds)
X_2 = average daily temperature (°F)
X_3 = number of working machines on a day
X_4 = number of employees working on a day
Y = the daily water usage (gallons)

	X_1	X_2	X_3	X_4	Y
1	7107	58.8	21	129	3067
2	6373	65.2	22	141	2828
3	6796	70.9	22	153	2891
4	9208	77.4	20	166	2994
5	14792	79.3	25	193	3082
6	14564	81	23	189	3898
7	11964	71.9	20	175	3502
8	13526	63.9	23	186	3060
9	12656	54.5	20	190	3211
10	14119	39.5	20	187	3286
11	16691	44.5	22	195	3542
12	14571	43.6	19	206	3125
13	13619	56	22	198	3022
14	14575	64.7	22	192	2922
15	14556	73	21	191	3950
16	18573	78.9	21	200	4488
17	15618	79.4	22	200	3295

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Regression Analysis: y versus x1; x2; x3; x4

X_1 = daily amount of production (in pounds)
X_2 = average daily temperature (°F)
X_3 = number of working machines on a day
X_4 = number of employees working on a day
Y = the daily water usage (gallons)

y = 6360 + 0.212 x1 + 13.9 x2 - 127 x3 - 21.8 x4

Predictor	Coef	SE Coef	T	P
Constant	6360	1314	4.84	0.000
x1	0.2117	0.04554	4.65	0.001
x2	13.869	5.160	2.69	0.020
x3	-126.69	48.02	-2.64	0.022
x4	-21.818	7.285	-3.00	0.011

S = 249.0 R-Sq = 76.7% R-Sq(adj) = 68.9%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	4	2448834	612209	9.88	0.001
Residual Error	12	743798	61983		
Total	16	3192632			

	X1	X2	X3	X4
Min	6373	39.5	19	129
Max	18573	81	25	206
Average	12900	64.85	21.47	182
St. Dev	3527	13.51	1.46	22

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Questions

- Is this a significant regression model?
- Explain the meaning of the regression model: how does each independent variable affect daily water usage? Do they increase or decrease the daily water usage? Are their effects significant?
- How much of the variability in daily water usage is explained by this model? Is this sufficient?
- The management aims to increase the production to a level of 20000 pounds, which may require the use of 25 machines and 250 employees. Can this model be used to predict the daily water usage in this case? Why or why not? Explain.

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Interpretation of r in multiple regression with k independent variables

$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$ coefficient of multiple determination

As you add more variables: $r^2 \uparrow$

But we lose one df for each additional parameter: not necessary a better model.

We need to check on MSE as well: $MSE = \frac{SSE}{n-k-1} \downarrow ?$

We can add new independent variables to explain Y. For example if we have n observations and n−1 variables:

$Y_j = \beta_0 + \beta_1 X_{1j} + \beta_2 X_{2j} + \ldots + \beta_{n-1} X_{n-1j}$ j = 1,...,n

⇒ an almost perfect fit as $r^2 = 1$, but MSE ↑ ∞.

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Evaluation criteria for "best"

r^2 is adjusted for the number of parameters in the model:

$r_a^2 = 1 - \frac{SSE/(n-k)}{SST/(n-1)}$ adjusted coefficient of multiple determination

Equivalent to minimizing S^2=MSE.

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Evaluation criteria for "best"

When p is the total number of independent variables. C_p statistics for k≤p:

$$C_p = \frac{(n-p-1)\mathrm{SSE}_{\mathrm{reduced}}(k)}{\mathrm{SSE}_{\mathrm{full}}(p)} - (n-2k-2)$$

A desirable model is then specified by a subset of predictors (regressors) for which C_p is small. The number of predictors in this set is k.

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How to choose the best regression model?

- Different criteria
  - Higher adjusted coefficient of multiple determination:

$$r_a^2 = 1 - \frac{SSE/(n-k)}{SST/(n-1)}$$

  - Equivalently lower MSE
  - Lower Mallows' C_p

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Evaluation criteria for "best"

All possible regressions: considerable computational effort.
k-candidate variables 2^k total equations

"best" subset of regressors…

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Problems in multiple regression

1. Multicollinearity occurs when two or more of the regressors are "highly" correlated. Then the estimates of the corresponding 's. have high variances.

   To detect multicollinearity:

   1. Inspection of sample correlations between x_j and x_k. for all j.k. If sample correlation $|r_{jk}| \approx 1$: suspect multicollinearity between x_j & x_k if more than 2 variables are correlated. this may not work.
   2. Let x_j be one variable that is suspected to have high correlation with one or more of the other x_k.

      Regression of x_j on x_k: If R^2 is "high" multicollinearity.
   3. F-test for regression is significant, but individual tests on β_i are not. Then suspect multicollinearity.
2. Influential observations in regression (problem for simple regression as well).
3. Autocorrelation (simple regression as well).

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Example: A Final Exam Question

- A worldwide organization wants to estimate the literacy ratio, LR, in any given settlement district, where LR is:

$$LR = \frac{number\ of\ people\ who\ can\ read\ \&\ write}{total\ population\ size}$$

- Their first step of analysis uses an economic indicator, EI, to generate rough predictions of literacy ratio. The economic indicator can take continuous values in scale (0,10), where 0 is the worst possible economic situation and 10 is the best.

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Example:
A Final Exam Question

$$r^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Regression Analysis: LR versus EI

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The regression equation is
LR = 0.622 + 0.0392 EI
Predictor       Coef    SE Coef       T       P
Constant     0.62206    0.02445   25.44   0.000
EI          0.039221   0.007856    4.99   0.000

S = 0.0885719   R-Sq = 24.2%   R-Sq(adj) = 23.2%

Analysis of Variance
Source           DF       SS       MS      F      P
Regression        1  0.19551  0.19551  24.92  0.000
Residual Error   78  0.61191  0.00784
Total            79  0.80742
```

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Categorical or dummy variables

- Environment
- Man/Woman/Children
- …

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Example: A Final Exam Question

- After analyzing the above regression model, and considering the rest of data they have, the statistical team of the organization suggested another regression model, which now indicates whether the settlement is an urban (city) or rural (country) area.

$$U - R = \begin{cases} 1 & \text{if urban} \\ 0 & \text{if rural} \end{cases}$$

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Example: A Final Exam Question

Regression Analysis: LR versus EI, U-R

The regression equation is

LR = 0.576 + 0.0356 EI + 0.0961 U-R

Predictor	Coef	SE Coef	T	P
Constant	0.57575	0.02232	25.80	0.000
EI	0.03565	0.00669	5.33	0.000
U-R	0.09613	0.01714	5.61	0.000

S = 0.0751069 R-Sq = 46.2% R-Sq(adj) = 44.8%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	0.37306	0.18653	33.07	0.000
Residual Error	77	0.43436	0.00564		
Total	79	0.80742			

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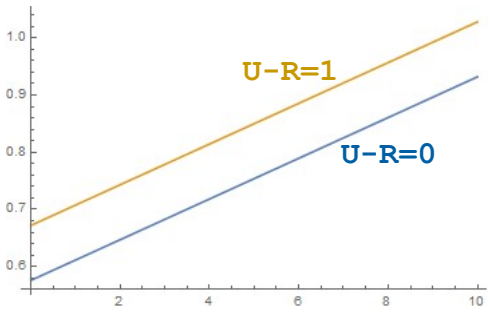
Example: A Final Exam Question

Regression Analysis: LR versus EI, U-R

The regression equation is

LR = 0.576 + 0.0356 EI + 0.0961 U-R

$$U-R=\begin{cases}1 & \text{if urban}\\ 0 & \text{if rural}\end{cases}$$



What does this equation mean?

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Example: A Final Exam Question

- What if we wanted to check the effects of continents (Europe, Asia, North America) on literacy ratio, LR?

$$C = \begin{cases} 0 & Europe \\ 1 & Asia \\ 2 & North\ America \end{cases}$$



$$LR = \beta_0 + \beta_{EI}\ EI + \beta_{U\text{-}R}\ U\text{-}R + \beta_C C + \varepsilon$$

Is the effect of being in Europe instead of Asia equal to the effect of being in Asia instead of North America?

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Example: A Final Exam Question

- What if we wanted to check the effects of continents (Europe, Asia, North America) on literacy ratio, LR?

$$E = \begin{cases} 1 & if\ in\ Europe \\ 0 & Otherwise \end{cases} \qquad N = \begin{cases} 1 & if\ in\ North\ America \\ 0 & Otherwise \end{cases}$$

$$A = \begin{cases} 1 & if\ in\ Asia \\ 0 & Otherwise \end{cases}$$

$$LR = \beta_0 + \beta_{EI}\ EI + \beta_{U\text{-}R}\ U\text{-}R + \beta_E E + \beta_A A + \beta_N N + \varepsilon$$

Do we need all three?

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Example: A Final Exam Question

- What if we wanted to check the effects of continents (Europe, Asia, North America) on literacy ratio, LR?

$$E = \begin{cases} 1 & if\ in\ Europe \\ 0 & Otherwise \end{cases} \qquad A = \begin{cases} 1 & if\ in\ Asia \\ 0 & Otherwise \end{cases}$$

$$LR = \beta_0 + \beta_{EI}\, EI + \beta_{U\text{-}R}\, U\text{-}R + \beta_E E + \beta_A A + \varepsilon$$

Europe	E=1, A=0
Asia	E=0, A=1
North America	E=0, A=0

- Meanings of β_E and β_A?
- What is the effect of being in Europe instead of Asia?

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